

MAT 21C Discussion 8

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§I Warm-Up

(Warm-up) If $\langle a, b \rangle$ is a unit vector, then for what values of a and b is the dot product

$$\langle 4, 3 \rangle \cdot \langle a, b \rangle = 4a + 3b$$

maximized? (Hint: think about the cosine formula for the dot product.)

Solution. The cosine formula says that

$$\langle 4, 3 \rangle \cdot \langle a, b \rangle = |\langle 4, 3 \rangle| |\langle a, b \rangle| \cos \theta,$$

where θ is the angle between the two vectors. Since $|\langle 4, 3 \rangle| = 5$ and $|\langle a, b \rangle| = 1$, the dot product is equal to $5 \cos \theta$. This is maximized when $\theta = 0$, which means that the two vectors are in the same direction. The unit vector pointing in the direction of $\langle 4, 3 \rangle$ is

$$\frac{\langle 4, 3 \rangle}{|\langle 4, 3 \rangle|} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle.$$

This is the first constrained optimization problem we solved analytically! In the warm-up, we

$$\text{optimized } 4a + 3b \text{ under the constraint } a^2 + b^2 = 1.$$

We still had to be a little clever to use the cosine formula for dot product. At the end of the quarter, we will learn a systematic method called **Lagrange multipliers** to solve this kind of problem without having to be clever.

§II Partial Derivatives

A two-variable function $f(x, y)$ has two partial derivatives, $f_x(x, y)$ and $f_y(x, y)$. They represent how fast $f(x, y)$ changes as we take a small step in the x -direction and y -direction, respectively. Consider the contour plot of $f(x, y)$ below.

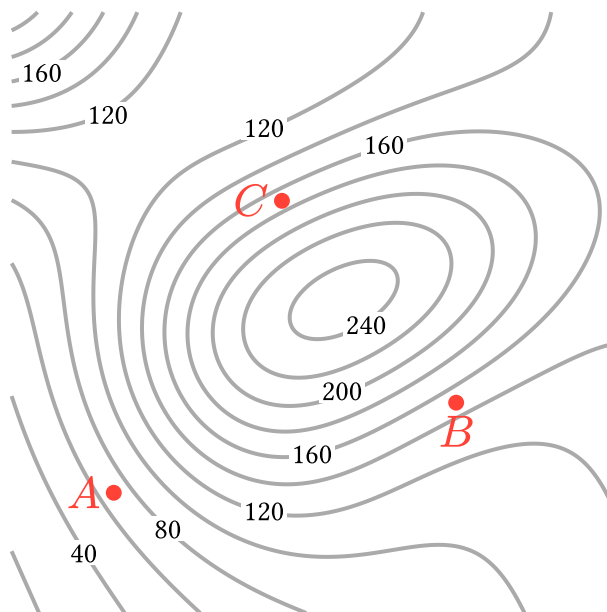


Figure 1: The contour plot of a function representing a hill. Along a contour line, the function has constant value labeled by the number on the line.

At point A, if we take a small step in the x -direction (towards the right), we will cross the contour line where the function value is 80. Looking at the surrounding contour lines, it looks like we are going from somewhere below 80 to somewhere above 80, so the function is increasing as we take the step, and we can conclude that $f_x > 0$ at the point A. Similarly, we can also see that $f_y > 0$ at the point A because we are also going up the hill in the y -direction.

Computationally, to find the derivative of $f(x, y)$ with respect to x , we treat y as a constant and vice versa. For example, if $f(x, y) = x^4 y^3$, then $f_x = 4x^3 y^3$ and $f_y = 3x^4 y^2$.

- The x -derivative of $x^4 1^3$ is $4x^3 1^3$.
- The x -derivative of $x^4 2^3$ is $4x^3 2^3$.
- The x -derivative of $x^4 \pi^3$ is $4x^3 \pi^3$.
- The x -derivative of $x^4 y^3$ is $4x^3 y^3$.

When taking the x -derivative, we treat y as a constant just like 1, 2, and π are constants.

Another notation for partial derivative is that

$$f_x(x, y) = \frac{\partial f}{\partial x}(x, y) \quad \text{and} \quad f_y(x, y) = \frac{\partial f}{\partial y}(x, y).$$

(Exercise)

1. Looking at the contour plot above. Find the signs of f_x and f_y at points B and C.
2. Let $f(x, y) = x^2 + y^2$, find f_x and f_y .
3. Let $f(x, y) = \ln(2x + 3y)$, find f_x and f_y .

Solution.

1. At point B, we are going down the hill in the x -direction and going up the hill in the y -direction. So $f_x < 0$ and $f_y > 0$ at point B.

At point C, we are going up the hill in the x -direction and going down the hill in the y -direction, so $f_x > 0$ and $f_y < 0$ at point C.

2. We have $f_x = 2x$ and $f_y = 2y$.
3. We have

$$f_x = \frac{2}{2x + 3y} \quad \text{and} \quad f_y = \frac{3}{2x + 3y}.$$

§III Chain Rule

Let $z(x, y)$ be a function of x and y , we have discussed that

$$\frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y}$$

represent how fast z changes as we take a small step in the x -direction and y -direction, respectively. Now, if we take a small step dx in the x -direction and a small step dy in the y -direction, the change in z is going to be the sum of

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

The same idea leads to the multivariate chain rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

when $x(t)$ and $y(t)$ are both functions of t . For example, if we have $z = x^2 + y^2$, $x = \cos t$, and $y = \sin t$, then

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= 2x \cdot (-\sin t) + 2y \cdot \cos t \\ &= 2 \cos t \cdot (-\sin t) + 2 \sin t \cdot \cos t \\ &= 0. \end{aligned}$$

This agrees with a more direct computation that $z = x^2 + y^2 = \cos^2 t + \sin^2 t = 1$ and so $dz/dt = 0$.

A more general version applies to partial derivative as well. Given $z(x, y)$, $x(u, v)$, and $y(u, v)$, we have

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}.$$

(Exercise)

1. Let $z = \ln(2x + 3y)$, $x = u + 3v$, and $y = u - 2v$, find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.
2. Let $z = x^4 y^3$, $x = 1 + \frac{t}{\sqrt{5}}$, and $y = 1 + \frac{2t}{\sqrt{5}}$, find $\frac{dz}{dt}$ at $t = 0$.
3. Let $\langle a, b \rangle$ be a unit vector and

$$z = x^4 y^3, \quad x = 1 + at, \quad y = 1 + bt.$$

Find $\frac{dz}{dt}$ at $t = 0$, and find the values of a and b that maximize $\frac{dz}{dt}$ at $t = 0$. Explain why this vector $\langle a, b \rangle$ is the direction of the “steepest ascent” of the function $z = x^4 y^3$ at the point $(1, 1)$.

Solution.

1. This problem can be easily done with substitution, we have

$$z = \ln(2(u + 3v) + 3(u - 2v)) = \ln(5u).$$

So we have

$$\frac{\partial z}{\partial u} = \frac{5}{5u} = \frac{1}{u} \quad \text{and} \quad \frac{\partial z}{\partial v} = 0.$$

We can definitely still do it with the chain rule as well. We have found in the previous exercise that

$$\frac{\partial z}{\partial x} = \frac{2}{2x + 3y} \quad \text{and} \quad \frac{\partial z}{\partial y} = \frac{3}{2x + 3y}.$$

We also have

$$\frac{\partial x}{\partial u} = 1 \quad \text{and} \quad \frac{\partial y}{\partial u} = 1$$

and so

$$\frac{\partial z}{\partial u} = \frac{2}{2x + 3y} \cdot 1 + \frac{3}{2x + 3y} \cdot 1 = \frac{5}{2x + 3y} = \frac{1}{u}.$$

Similarly, we have

$$\frac{\partial x}{\partial v} = 3 \quad \text{and} \quad \frac{\partial y}{\partial v} = -2$$

and so

$$\frac{\partial z}{\partial v} = \frac{2}{2x + 3y} \cdot 3 + \frac{3}{2x + 3y} \cdot (-2) = 0.$$

Sure enough, we get the same answers using both methods.

2. Substitution would not make progress to solve the problem. This is where the chain rule comes in handy. We computed in the previous part that

$$\frac{\partial z}{\partial x} = 4x^3y^3 \quad \text{and} \quad \frac{\partial z}{\partial y} = 3x^4y^2,$$

We also have

$$\frac{\partial x}{\partial t} = \frac{1}{\sqrt{5}} \quad \text{and} \quad \frac{\partial y}{\partial t} = \frac{2}{\sqrt{5}}.$$

So we have

$$\frac{dz}{dt} = 4x^3y^3 \cdot \frac{1}{\sqrt{5}} + 3x^4y^2 \cdot \frac{2}{\sqrt{5}}.$$

Instead of expanding x and y into expressions in terms of t and plugging in $t = 0$, it is simpler to the values of x and y at $t = 0$ and then plug them into the expression above. At $t = 0$, we have

$$x = 1 \quad \text{and} \quad y = 1,$$

so

$$\frac{dz}{dt} = 4\frac{1}{\sqrt{5}} + 3\frac{2}{\sqrt{5}} = \frac{10}{\sqrt{5}}.$$

We can write dz/dt completely in terms of t first then plug in $t = 0$. Doing this, we have

$$\begin{aligned}\frac{dz}{dt} &= 4\left(1 + \frac{t}{\sqrt{5}}\right)^3 \left(1 + \frac{2t}{\sqrt{5}}\right)^3 \cdot \frac{1}{\sqrt{5}} + 3\left(1 + \frac{t}{\sqrt{5}}\right)^4 \left(1 + \frac{2t}{\sqrt{5}}\right)^2 \cdot \frac{2}{\sqrt{5}} \\ &= 4\frac{1}{\sqrt{5}} + 3\frac{2}{\sqrt{5}}\end{aligned}$$

which is of course the same answer but has a lot to write. Knowing to compute x and y at $t = 0$ and plugging them in is a very useful trick.

3. In general, we have

$$\frac{\partial x}{\partial t} = a \quad \text{and} \quad \frac{\partial y}{\partial t} = b,$$

and so,

$$\frac{dz}{dt} = 4x^3y^3a + 3x^4y^2b.$$

At $t = 0$, we have $x = 1$ and $y = 1$, so

$$\frac{dz}{dt} = 4a + 3b.$$

To maximize this for unit vector $\langle a, b \rangle$, we have seen in the warm-up problem that the maximum is achieved when $\langle a, b \rangle$ points in the same direction as $\langle 4, 3 \rangle$, which means that

$$\langle a, b \rangle = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle.$$

This vector is the direction of the steepest ascent because

$$\frac{dz}{dt} = 4a + 3b$$

is telling us how fast z is changing as we take a small step in the direction of $\langle a, b \rangle$ starting from $(1, 1)$. Maximizing this quantity is equivalent to answering the question “which direction should we take a small step from $(1, 1)$ to increase the value of z the most?” Hence, the name steepest ascent.

We can even generalize this argument to see that the **gradient vector**

$$\left\langle \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \right\rangle = \hat{i} \frac{\partial z}{\partial x} + \hat{j} \frac{\partial z}{\partial y}$$

always points in the direction of the steepest ascent of the function z at any point.